

Practical - 3

Practical Title: Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

Problem Definition: Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.

F. Theory

1. How does the layout adapt for mobile, tablet, and desktop screens?

Answer: A layout adapts across mobile, tablet, and desktop screens using fluid grids, flexible images, and breakpoints (typically under 600px for mobile, 601px–1024px for tablets, and 1025px+ for desktops). Elements reflow, resize, or stack vertically to fit the available screen space.

(https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Responsive_Design)

2. Which layout technique is used and why?

Answer: Common web design layout techniques include **CSS Flexbox**, **CSS Grid**, and traditional **Float** or **Responsive** structures. Because your specific project or graphic was not provided, the correct technique depends entirely on the design goal.

(https://www.w3schools.com/html/html_layout.asp)

3. Are typography, spacing, colors, and contrast readable?

Answer: Yes, typography, spacing, colors, and contrast directly determine whether text is readable. When properly balanced, they prevent eye strain and make content easy to scan. When poorly configured, text becomes difficult or impossible to read.

(<https://guidelines.india.gov.in/activity/the-impact-of-typography-on-web-design-a-comprehensive-guide-to-aesthetics-and-readability/>)

4. Is the framework used consistently without unnecessary inline CSS?

Answer: `style=""` attributes. While they look like inline styles in the markup, they compile into clean external stylesheets and maintain design consistency without unnecessary true inline CSS.

(https://www.reddit.com/r/Frontend/comments/rk3lr4/are_css_frameworks_really_necessary/)

G. Implementation/ Output/Result :

- CSS Selectors Used:

*, body, header, nav, main, section, article, footer, div,
h1, h2, h3, h4, p, ul, li, a, img,
form, label, input, textarea, select, button,
table, tr, td, hr, small, i, marquee,
.class, #id,
:hover, :focus, :first-child, :last-child,
input[type="text"], input[type="email"], input[type="password"],
input[type="radio"], input[type="checkbox"],
input[type="submit"], input[type="reset"],
@media

- CSS Properties Used:

margin, padding, box-sizing,
display, width, height, max-width,
background, background-size, background-position, background-repeat,
color,
font-family, font-size, font-weight,
text-align, text-decoration, line-height,
border, border-left, border-top, border-radius,
box-shadow,
position,

justify-content, align-items,

flex-direction, flex-wrap,

gap,

grid-template-columns,

list-style,

cursor,

transition,

transform,

outline,

resize,

object-fit,

overflow

H. Evidence of Work (Screenshots)

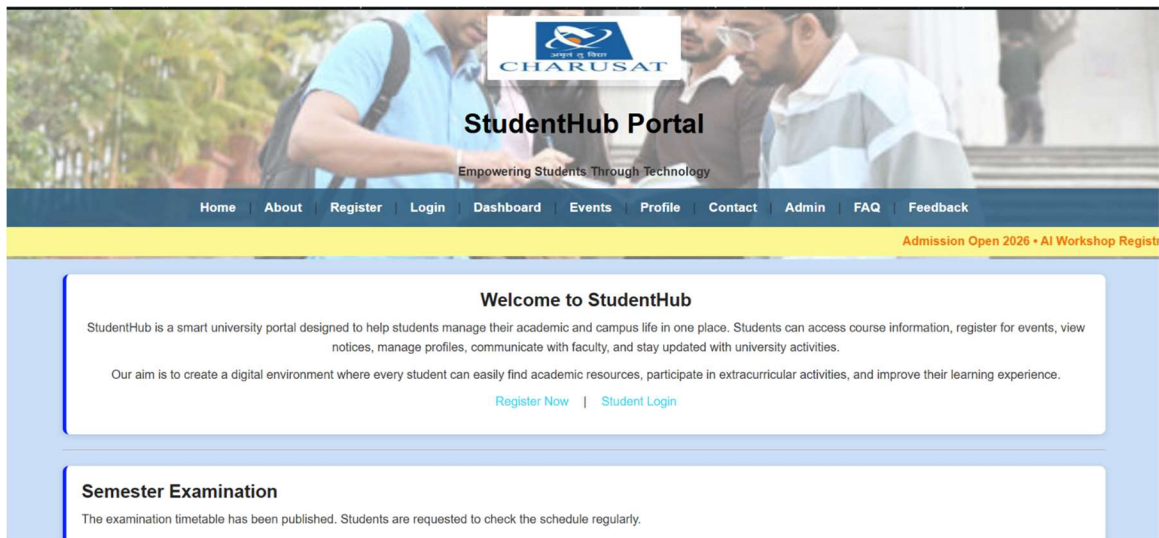


Figure 3.1(Home page)

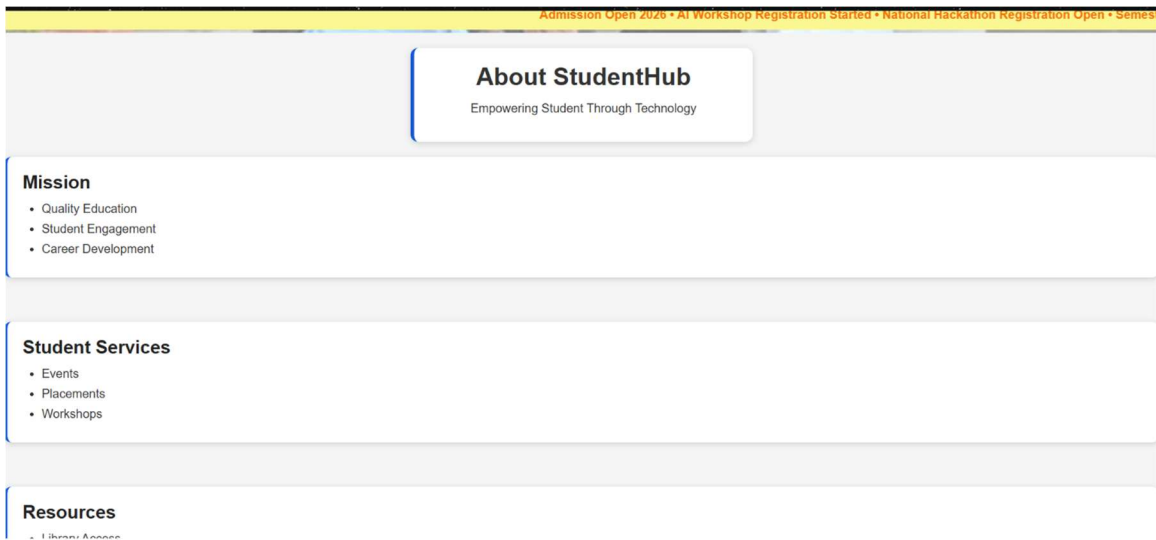


Figure 3.2(About page)

The screenshot shows the 'StudentHub Portal' registration page. The header is blue with the text 'StudentHub Portal', 'Create Your Account', and 'Already have an account? [Login Here](#)'. Below this is a white registration form titled 'Registration Form'. The form contains the following fields: 'Full Name' (text input), 'Email' (text input), 'Mobile' (text input), 'Password' (text input), 'Confirm Password' (text input), 'Date of Birth' (date picker with format 'dd-mm-yyyy'), and 'Gender' (radio buttons for 'Male', 'Female', and 'Other').

Figure 3.3(Registration page)

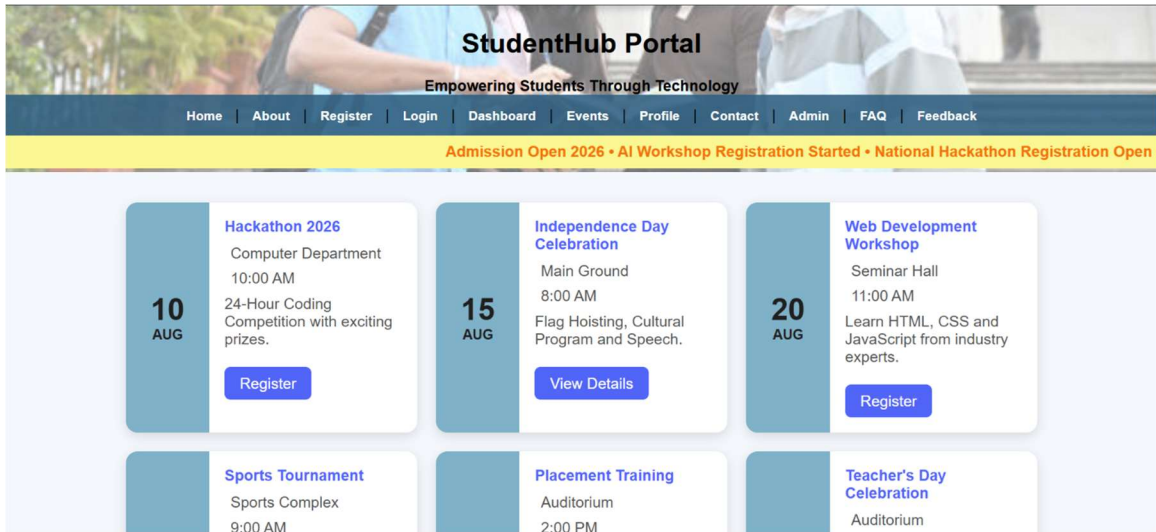


Figure 3.4(Event page)

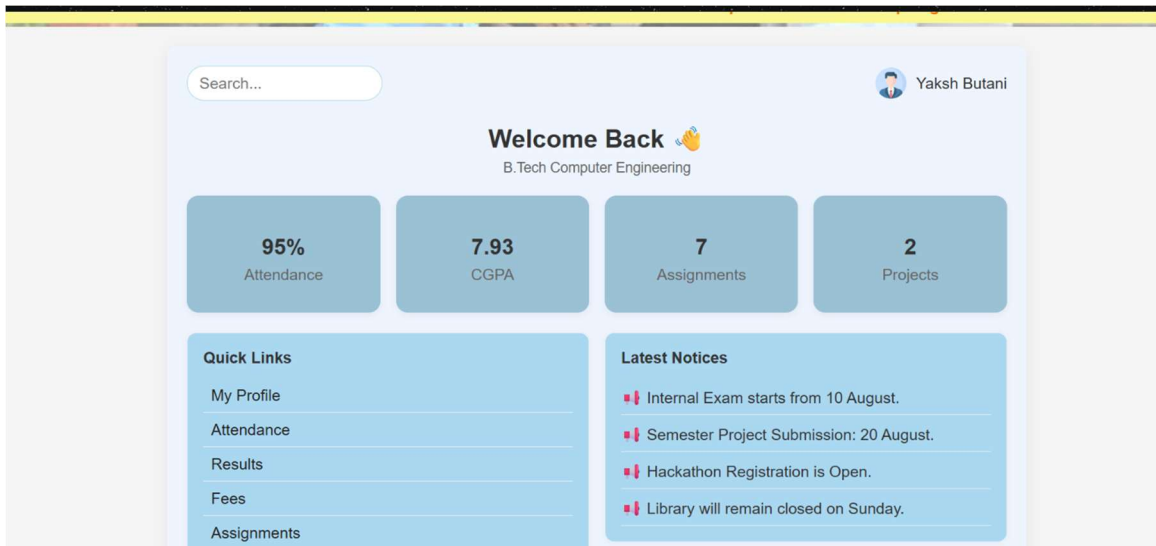


Figure 3.5(Dashboard page)

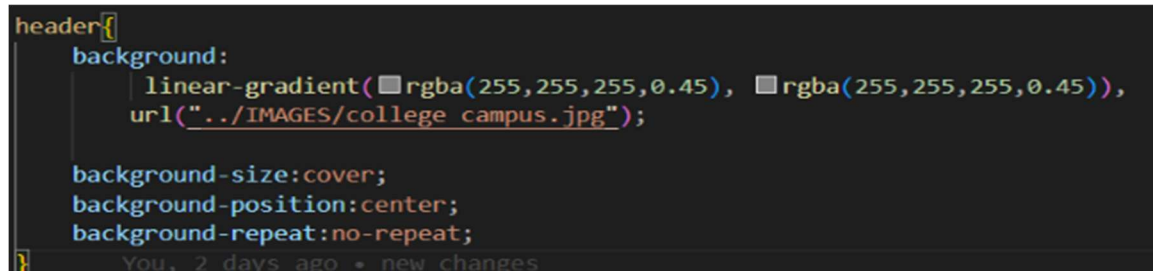
I. GitHub Repository Link:

<https://github.com/yakshu205/StudentHub.git>

J. Challenges Faced:

When I was styling the html files , I had no idea about how I put image behind the text and blur it . then I told ChatGPT that I want to put image behind text at header which concept or styling function I should use , give syntax and opacity code and gradient code like (255,255,255,0.20);

It gave me function that name is linear gradient and also I learned padding , card , form, Flex , grid, styling , box , responsive etc.. concept.

A screenshot of a code editor showing CSS code for a header. The code is as follows:

```
header{  
  background:  
    linear-gradient(rgba(255,255,255,0.45), rgba(255,255,255,0.45)),  
    url("../IMAGES/college campus.jpg");  
  
  background-size:cover;  
  background-position:center;  
  background-repeat:no-repeat;  
}
```

The code is highlighted with a dark background and light text. At the bottom of the screenshot, there is a status bar that says "You, 2 days ago • new changes".

K. Learning Outcomes:

I will design a professional, responsive, and reusable UI layout.

Faculty Remarks: (Keep this blank for faculty)